

VPN vs ZTNA – QA Comparison

A traditional VPN and a Zero Trust Network Access (ZTNA) solution both provide secure access to resources, but their access models are different. From a QA perspective, this difference changes what should be tested.

A VPN normally creates an encrypted tunnel between the user device and a VPN gateway. After successful authentication, the user may receive broader network-level access to internal resources. QA testing therefore focuses on network connectivity, tunnel establishment, routing, DNS, internal host reachability, firewall rules, and whether unauthorized network resources can be reached. Wireshark can also be used to verify the TCP connection and encrypted tunnel traffic.

ZTNA follows a least-privilege, application-focused model. Instead of giving the user general access to an internal network, it provides access only to applications or resources allowed by identity and policy. QA testing should therefore focus more on application access and policy decisions. Test cases should verify user identity, authentication, MFA, device or posture requirements where applicable, application-level authorization, session and token validation, and correct handling of unauthorized requests. A user who is authenticated but not authorized for an application should be denied, commonly with an appropriate HTTP status such as 403.

For VPN testing, the main question is: “Can the user securely reach the network?” For ZTNA testing, the main question is: “Can this specific user securely reach only the resources they are authorized to use?” This makes ZTNA testing more identity- and policy-driven than traditional network connectivity testing.